Example 1.33 In a residential area with 600 families $\frac{3}{5}$ owned scooter, $\frac{1}{3}$ owned car, $\frac{1}{4}$ owned bicycle, 120 families owned scooter and car, 86 owned car and bicylce while 90 families owned scooter and bicylce. If $\frac{2}{15}$ of families owned all the three types of vehicles, then find (i) the number of families owned atleast two types of vehicle. (ii) the number of families owned no vehicle.

Solution Let S, C and B represent sets of families who owned Scooter, Car and Bicycle respectively.

Given, $n(\mathrm{U}) = 600 \quad n(S) = \frac{3}{5} \times 600 = 360$

$n(C) = \frac{1}{3} \times 600 = 200 \quad n(B) = \frac{1}{4} \times 600 = 150$

$n(S \cap C \cap B) = \frac{2}{15} \times 600 = 80$

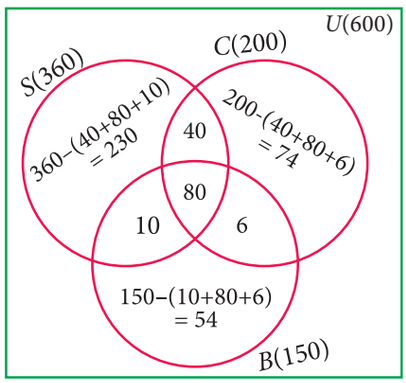


Fig.1.38

From Venn diagram,

(i) The number of families owned atleast two types of vehicles = 40+6+10+80 = 136

(ii) The number of families owned no vehicle

$= 600 - (\text{owned atleast one vehicle})$

$= 600 - (230 + 40 + 74 + 6 + 54 + 10 + 80)$

$= 600 - 494 = 106$

Example 1.34 In a group of 100 students, 85 students speak Tamil, 40 students speak English, 20 students speak French, 32 speak Tamil and English, 13 speak English and French and 10 speak Tamil and French. If each student knows atleast any one of these languages, then find the number of students who speak all these three languages.

Solution Let A, B and C represent sets of students who speak Tamil, English and French respectively.

Given, $n(A \cup B \cup C) = 100,\ n(A) = 85,\ n(\mathrm{B}) = 40,\ n(\mathrm{C}) = 20,$
$n(\mathrm{A} \cap B) = 32,\ n(\mathrm{B} \cap C) = 13,\ n(\mathrm{A} \cap C) = 10\,.$

We know that,

$$n(A \cup B \cup C) = n(A) + n(B) + n(\mathrm{C}) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$$

$$100 = 85 + 40 + 20 - 32 - 13 - 10 + n(A \cap B \cap C)$$

Then, $n(A \cap B \cap C) = 100 - 90 = 10$

Therefore, 10 students speak all the three languages.

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